

5CS037 – Concepts & Technologies of AI

Assignment - I - Statistical Interpretation and Exploratory Data
Analysis.

Analysis of the Human Development Index (HDI):

A Data - Driven Exploration of Global and Regional Development
Patterns.

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1.Introduction:

HDI stands for **Human Development Index**. It is a blend statistic developed by the United Nations Development Programme to measure and compare the average achievements in three basic human development:

1. Health.
2. Education.
3. Standard of living.

The main objective of analysis human development is observe all over the world and we should identify the pattern of human development .The research motive to analyze how the education ,income ,health and living stander indicate the overall infulance the value of HDI and show variation between high ,medium and low human development .

The scope of this report will be limited to the analysis of the available data on the HDI in the selected countries and regions within a fixed time span. Methodological note: The scope is limited as the analysis is more on the observations rather than the formulation of policies. The analysis will be utilized in deriving the global patterns on human development through the use of statistical analysis techniques.

1. Problem 1A – Single year HDI Exploration

Methods / Approach

In this analysis, the HDI of a country within a single year, namely 2021/2022, was considered using information from the UNDP Human Development Reports. Handling the data meant only including countries with full information on HDI and its major constituents: life expectancy, education, and GNI per capita variables. Descriptive statistics were performed, country categories were classified into the four categories used in UNDP HDI classifications, and ranking was performed. The analysis was performed to test the relationship between HDI variables and their constituents, including health, education, and income.

Key Results

- **Global Overview:** The global average HDI for the most recent year was 0.687, indicating a medium level of human development.
- **HDI Tier Distribution:** The results highlight strong global disparities:
 - **Very High HDI:** About 25% of countries (e.g., Australia, Austria, Argentina) fall into this category.
 - **High HDI:** Approximately 30% of countries (e.g., Antigua and Barbuda, Azerbaijan) are in the high development tier.
 - **Medium and Low HDI:** The remaining ~30% of countries (e.g., Bangladesh, Afghanistan) fall into medium and low development categories.
- **Top and Bottom Performers:**
 - **Highest HDI:** Australia (0.946) and Austria (0.926) show very high development, supported by high life expectancy and strong income levels.
 - **Lowest HDI:** Afghanistan (0.462), Bangladesh (0.518), and Angola (0.591) exhibit low development, marked by shorter life expectancy, lower educational attainment, and low income.
- **Regional Disparities:** Clear regional inequalities were observed, with high-income European countries averaging an HDI of ~0.82, while lower-income countries average around ~0.52.

Interpretation and discussion

The analysis of the HDI for a single year indicates disparities in levels of development across different countries. A country with a higher HDI has better levels of health, education, and income, while countries with a low HDI are however faced with development problems.

2. • Problem 1B – HDI Trend Analysis (2020–2022)

Methods / Approach

The focus is on the route of human development in the global arena for the critical years of 2020 to 2022, which include the period of the highest intensity of the COVID pandemic and the early stages of recovery. The purpose is to notice trends on a global arena. This study obtained data on the HDI for the period 2020 to 2022 through UNDP publications. This data was presented on a country by country and year by year basis, using HDI, as well as the remaining important parameters: life expectancy, education, and income. In analyzing these parameters, it was sought to establish global, continental, and HDI trends, effects of the covid 19 pandemic, as well as the component that affected changes in the HDI.

Key Results

Global Decline and Slow Recovery:

Global HDI declined in 2020 and 2021 due to Covid 19 the first decline ever recorded. Although HDI improved in 2022, it has not yet fully returned to prepandemic levels.

Unequal Impact Across Countries:

Developing and low income countries were affected and recovered slowly, while high income countries showed faster recovery.

Health as the Main Factor:

The largest decline in HDI was caused by reduced life expectancy due to covid 19 deaths, affecting both rich and poor countries.

Education Disruption:

Long school closures slowed or reversed education progress, especially in South Asia and Latin America.

Income Changes:

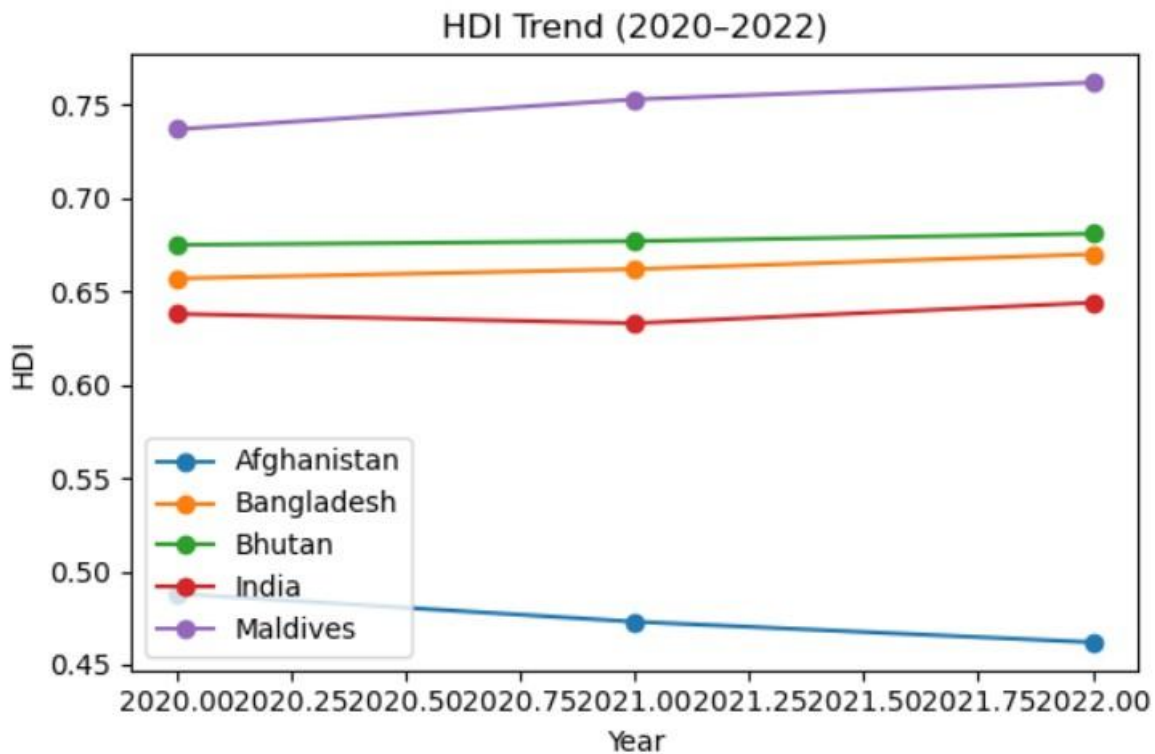
Incomes dropped sharply in 2020 due to job losses. Recovery occurred in 2021–2022, mainly in resource-rich and industrialized countries, while poorer nations lagged behind.

HDI Category Movement:

Most countries remained in the same HDI category. Some countries temporarily fell from higher HDI levels in 2021 but showed improvement again in 2022.

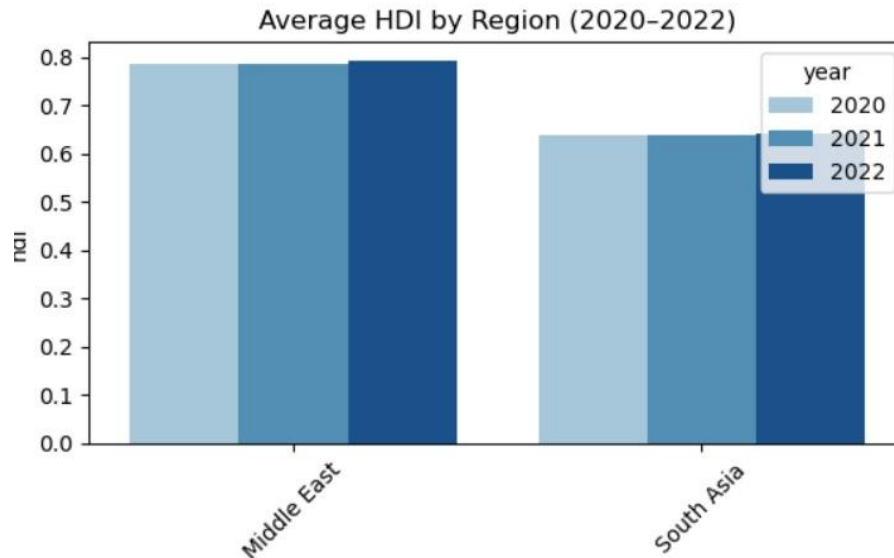
Visualizations and Tables

1.South Asia HDI Trends (Line Chart)



The chart displays HDI changes for five South Asian countries over three years. Maldives shows a stable increase in HDI, while Afghanistan's HDI declines. Bangladesh, Bhutan, India, and the Maldives show relatively stable or slight improvements in their HDI values.

2.Regional HDI Comparison (Grouped Bar Chart)



Comparison of HDI in Different Regions

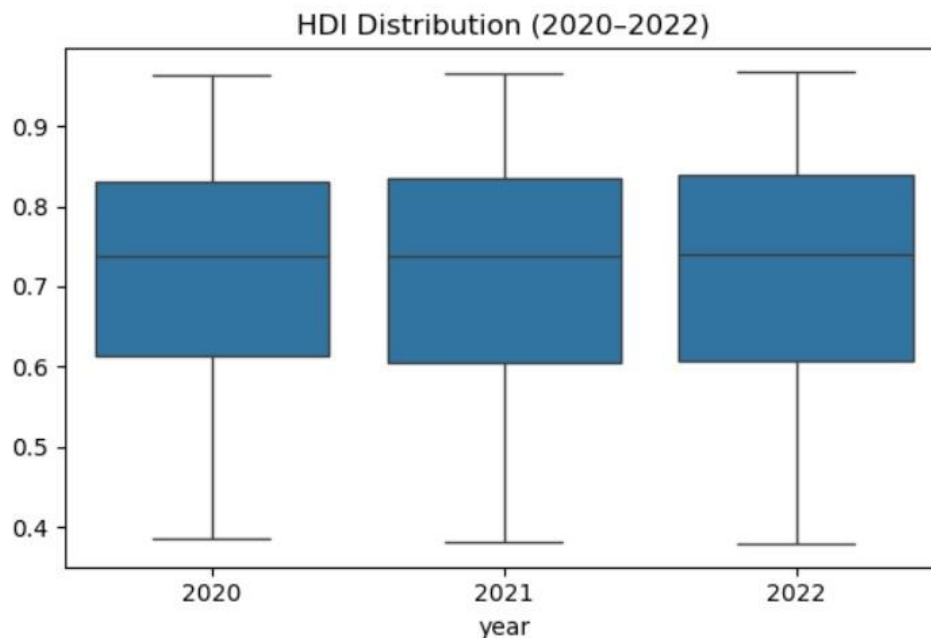
This graph shows Human Development Index values divided into regions across three successive years 2020, 2021, and 2022. It allows observing differences between the regions and their dynamics over the three years.

Points of interest: The Middle East region maintains its position well above the South Asian one in all three years. It has an HDI close to 0.79 against 0.64 in South Asia. Thus, there is a difference of about 0.15 points, which is quite a big number for the HDI indicator.

Implications: Pay attention to how flat the bars are in each of the regions 2020, 2021, and 2022 are almost identical. Hence, HDI does not fluctuate in the short term very much. Parameters like life expectancy, education level, and GDP per capita develop slowly; hence, no event of the global scale or policy decision made in a certain year will show up as a sharp increase or decrease. In fact, what we should focus on here is not year-on-year dynamics but the structural difference

between the regions.

3.Global HDI Distribution



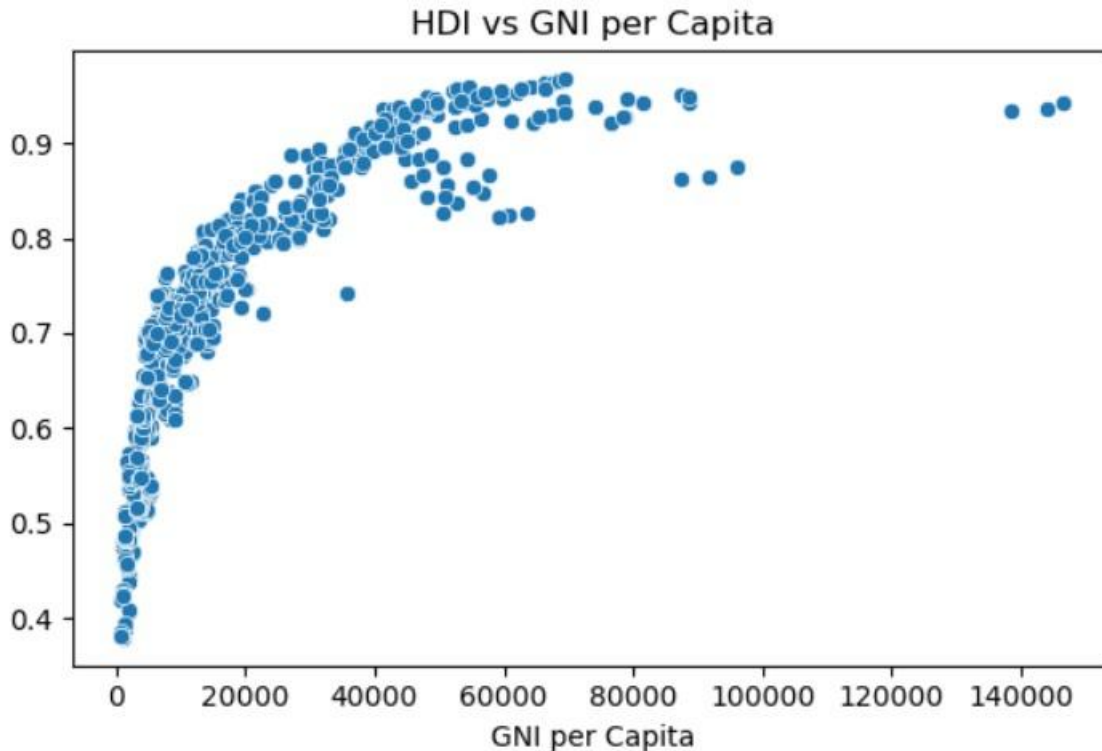
Spread of HDI for the Three Years

The following box plot allows us to go beyond averages and take a look at the distribution of HDI for countries within a particular year, not just the average country.

Observations: The shape is pretty much identical from 2020 to 2022. For each year, the interquartile range (the box) spans between roughly 0.61 and 0.84, with the median at 0.73 – 0.74. Whiskers range from 0.38 to 0.97, and this range is quite consistent throughout all three years.

Implications: As you can guess from the consistency of the shape in each year, what follows from that is something that was mentioned multiple times before – HDI does not change that often. The distribution and the laggards stay almost the

Scatter Plot – HDI vs GNI per Capita (2020–2022)



Comparison Between HDI and GNI Per Capita

Observation: There is a direct correlation between HDI and GNI per capita on the scatterplot below. As one increases, the other increases as well. Most of the nations are clustered in the low and mid income category, while the few rich nations are placed towards the right of the plot as outliers.

Insight: It proves that the income of a nation is a powerful determinant of its level of human development. However, it's not everything because the curve is becoming flatter at the higher income category. It means that the increase of income does not contribute to an increase in HDI at a certain point of development. Besides, there are some low income nations with high HDI figures, which is why we can consider the significance of non-monetary determinants such as education.

Interpretation and Discussion

The 2020–2022 period was a major shock to global human development, affecting more than just health. Covid 19 caused setbacks in all HDI components like education, health, and income threatening future human capabilities. Recovery varied widely: wealthier nations with better economic resources and vaccine access bounced back faster, while poorer countries suffered prolonged health, education, and economic challenges, widening existing development gaps.

3. Problem 2 – Advanced HDI Exploration

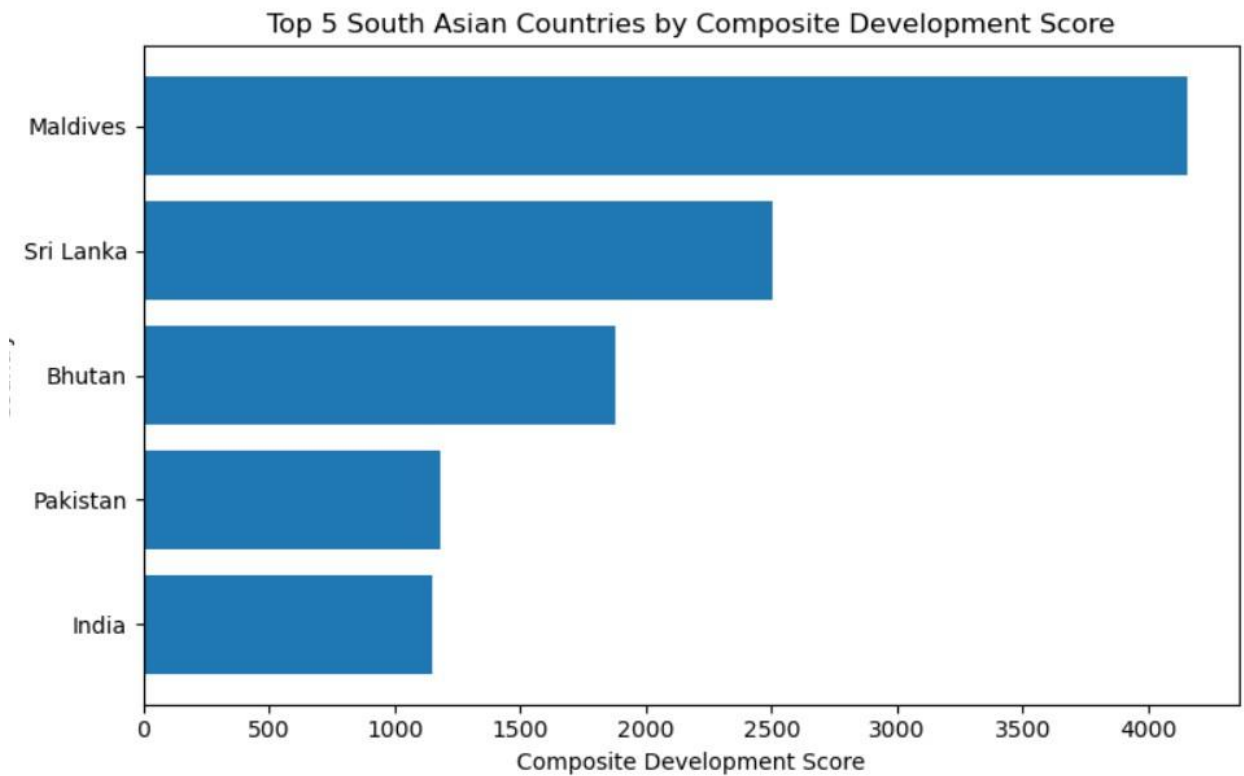
Methods / Approach

The analysis utilized the HDI dataset, focusing on South Asian countries. Indicators include HDI, life expectancy, GNI per capita, and gender development. Composite scores were developed to compare countries, and correlation analysis was performed by utilizing the IQR method to find relationships between variables and detect outliers, supported by visualizations.

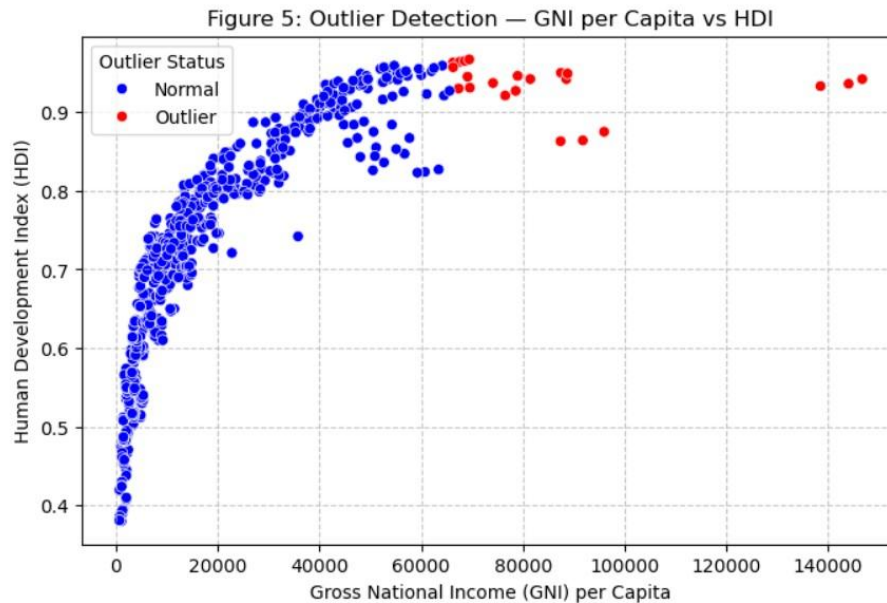
Key Results

- HDI levels showed clear differences among South Asian countries.
- Higher life expectancy and gender development were strongly associated with higher HDI.
- Some countries achieved relatively high HDI despite lower GNI per capita.
- Outlier analysis identified countries where income did not translate effectively into human development.

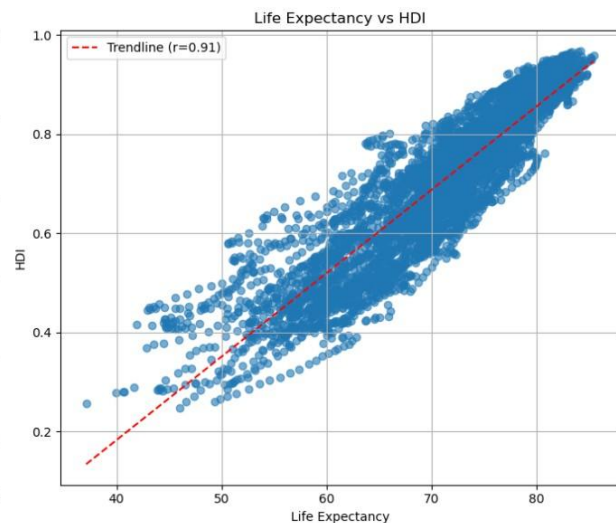
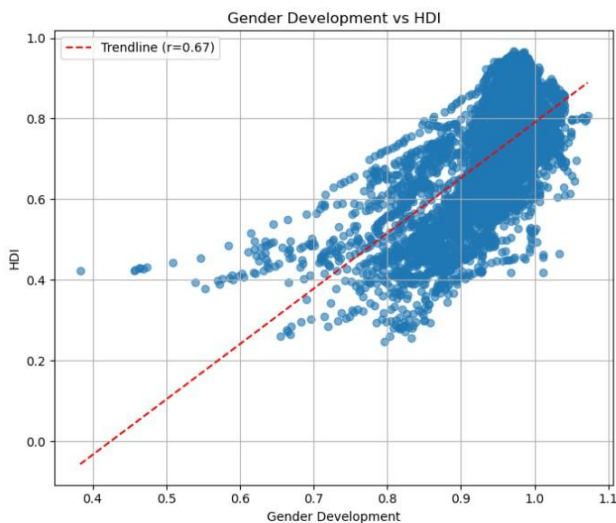
Visualizations and tables



The bar chart ranks the top 5 South Asian countries by composite development score. Maldives leads by a large margin, indicating the highest overall development. Sri Lanka follows then Bhutan. Pakistan and India have the lowest scores among the five showing comparatively lower composite development levels.



Correlation HDI vs Gender Development: 0.673
Correlation HDI vs Life Expectancy: 0.909



Outlier Detection GNI per Capita vs HDI

- Observation:** This chart flags specific countries as statistical outliers (shown in red) rather than just visual exceptions. Most countries follow the expected curve in blue — sharp gains in HDI at low income, flattening out as income rises. The red points break that pattern in two ways: a cluster around \$50,000–\$95,000 GNI with HDI sitting noticeably *lower* than nearby blue points (around 0.83–0.90), and a separate group near \$140,000+ GNI that stands apart simply because of how far out their income sits, even though their HDI (~0.93–0.94) isn't unusual.

- **Insight:** Formal outlier detection backs up what the eye already suspects — a handful of high-income nations don't convert their wealth into development as efficiently as expected, likely due to resource-driven income (oil-rich economies, for instance) that doesn't spread evenly into health and education. Meanwhile, the ultra-high-income outliers show that extreme wealth doesn't push HDI proportionally higher either — confirming, once again, that the income-to-development relationship has a ceiling.

Correlation Check: Gender Development and Life Expectancy vs HDI

- **Observation:** Life Expectancy correlates very strongly with HDI ($r = 0.91$), with points hugging tightly around the trendline. Gender Development also correlates positively but much more loosely ($r = 0.67$), with a noticeably wider scatter around its trendline.
- **Insight:** This makes sense given how HDI is built — life expectancy is one of its direct components, so the tight relationship is almost guaranteed by design. Gender Development isn't a direct input, so its weaker, messier correlation tells us something more real: a country can reach a fairly high HDI score while still carrying significant gender gaps. In other words, overall development doesn't automatically mean equal development across genders — that's a separate axis worth tracking on its own.

Interpretation and Discussion

From these graphs, we can show that HDI improves as a result of increased life expectancy, income, and gender equality. In the case of South Asia, there are nations with a lack of income yet a relatively improved HDI. The outliers coincide with nations possessing a relatively high income yet an improved HDI, indicating a lack of synergy within development.

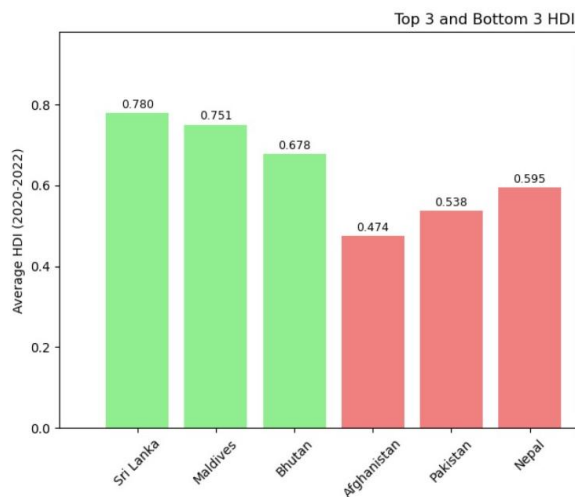
Problem 3 – Comparative Regional Analysis: South Asia vs Middle East

Methods / Approach

This research compares the growth of the human development of South Asia and the Middle East during the period 2020 to 2022. Data gathered from the HDI was utilized to establish the growth of the regions and the average statistics of the regions on the levels of development. The top and lowest developing country is also established. Other statistics, such as the average lifespan of the populations, female growth, and income, are also compared. Graphs and charts are applied to illustrate the relationship.

Key Results

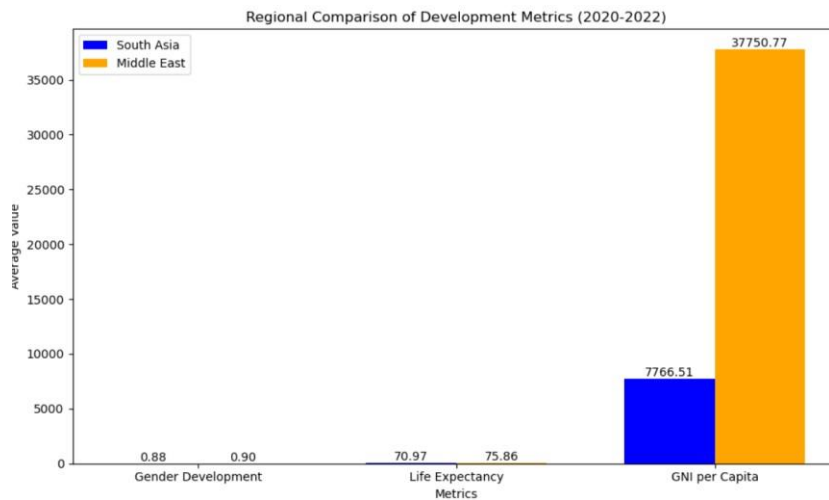
- The middle east shows a higher average HDI than South Asia during 2020 to 2022 indicating stronger overall human development.
- South Asia has a lower mean HDI and higher coefficient of variation, suggesting greater inequality and uneven development across countries.
- Top HDI performers in the middle east significantly outperform bottom performers, while South Asia shows a tighter but lower HDI range.
- Regional averages for life expectancy, gender development, and income are consistently higher in the middle east.
- Correlation analysis shows strong positive relationships between HDI and both life expectancy and gender development in both regions, with stronger associations in the middle east.



Top 3 and Bottom 3 HDI – on GNI-HDI Gap

- **Observations:** The selection of these six countries isn't based on the HDI ranking; instead, it is based on the discrepancy between actual HDI and predicted HDI based on the income level of these countries. Sri Lanka, Maldives and Bhutan have the maximum positive discrepancy; this implies that their HDI (0.78, 0.75 and 0.68 respectively) is greater than what their respective GNI level suggests. On the other hand, the minimum positive discrepancy belongs to Afghanistan, Pakistan and Nepal; their HDI levels (0.474, 0.538, 0.595) fall significantly short of predicted HDI level.
- **Insights:** The insight provided here is a lot more interesting than the ranking itself. It gives us an idea of those countries which are "punching above their weight;" i.e., the three countries in question have probably made relatively better investments in terms of health and education compared to their income level. As for the countries with lower discrepancy, it means that they have problems which prevent further development despite their income level.

Metric Comparisons: South Asia vs Middle East (2020 – 2022)

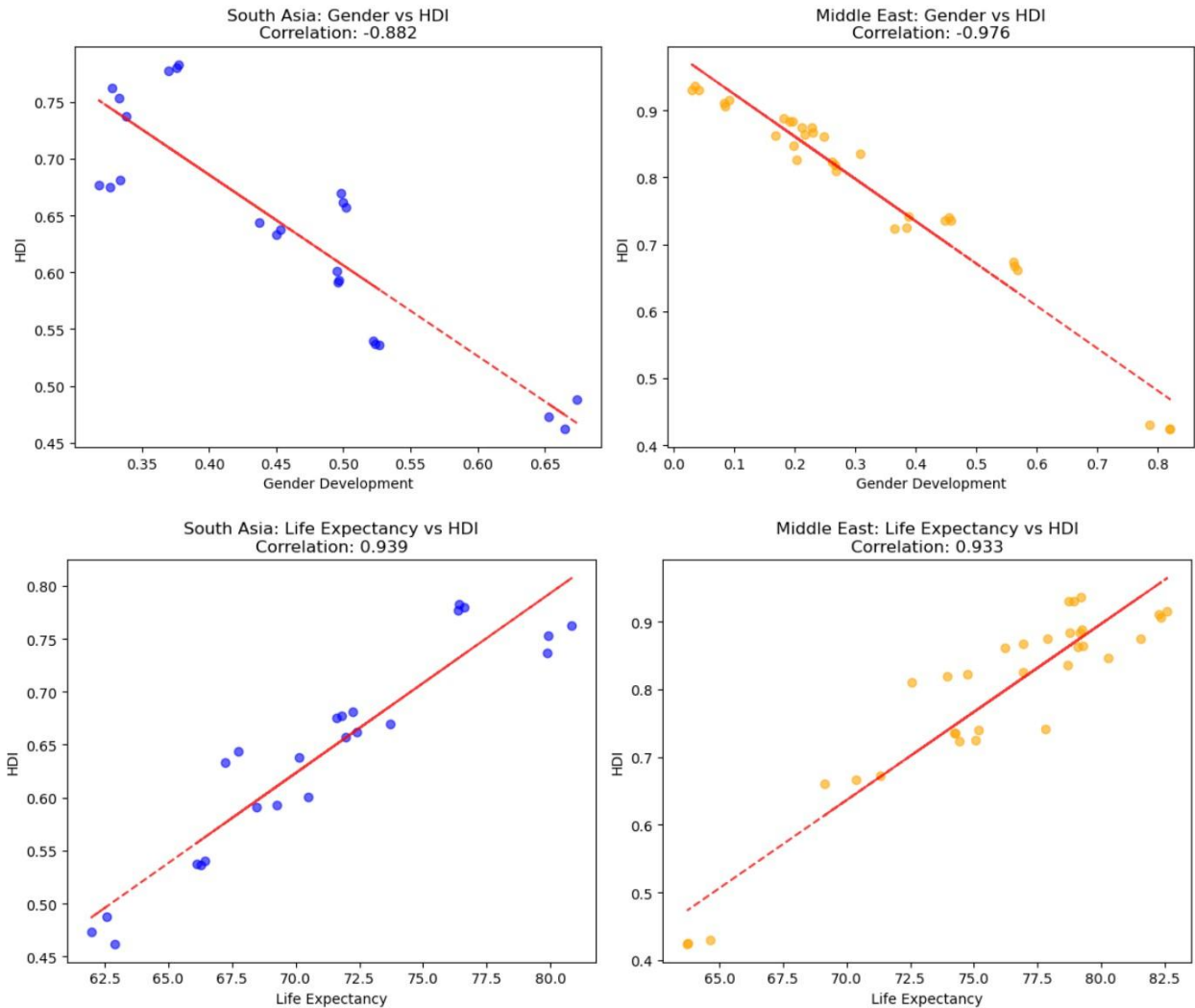


This chart compares three development metrics — Gender Development Index, Life Expectancy, and Gross National Income (GNI) per Capita — between South Asia and the Middle East.

Insight:

- **GNI per Capita** is significantly higher in the Middle East (above 35,000) than in South Asia (below 10,000).
- **Life Expectancy** is slightly higher in the Middle East.
- **Gender Development Index** values are low in both regions, with minimal difference.

Scatter Plots – HDI vs Gender Development & Life Expectancy (South Asia, 2020– 2022)

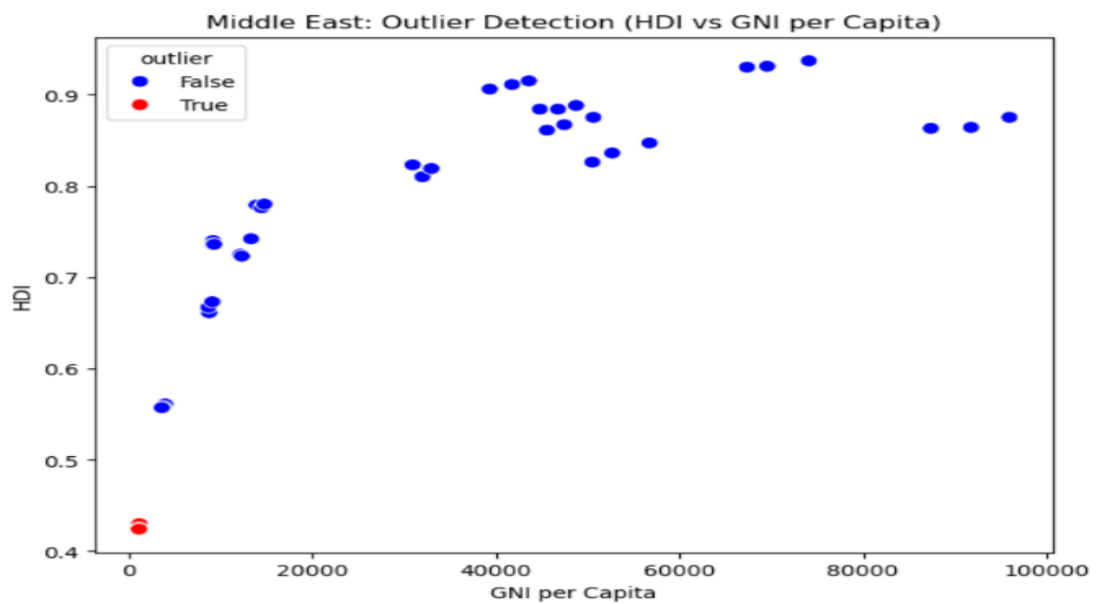
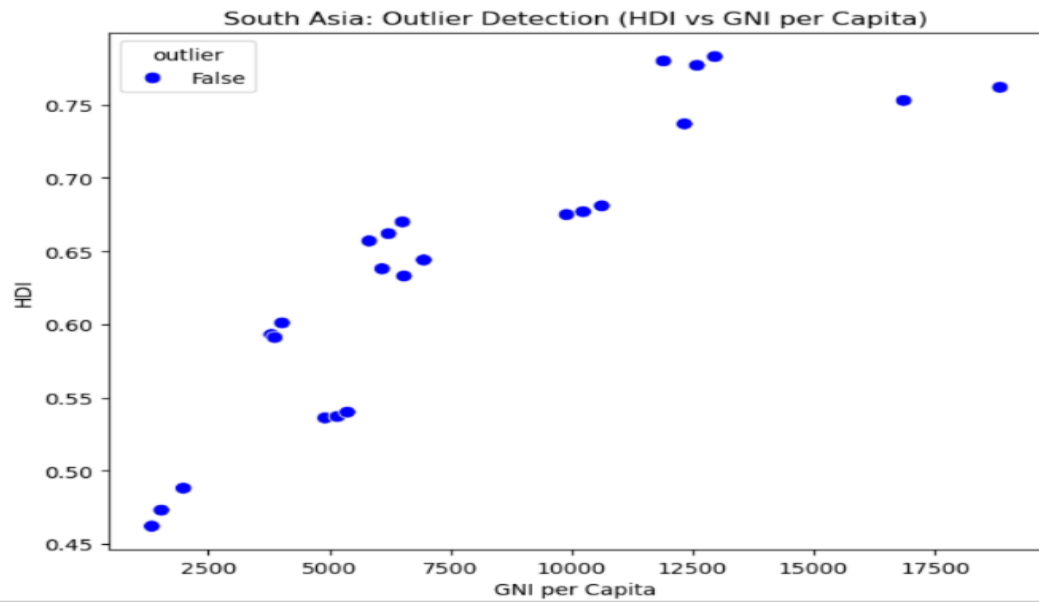


South Asia: Gender Development vs. HDI vs. Life Expectancy vs. HDI

- Gender Development vs. HDI ($r = -0.882$):** This is a strong *negative* relationship — as the gender metric rises, HDI falls. That's the opposite of what you'd expect if this were a standard "Gender Development Index" (where higher = more parity = should track with HDI). The negative slope suggests this axis is more likely measuring a *gap or inequality index* (e.g., something like the Gender Inequality Index), where higher values mean *more* inequality — which would explain why it drags HDI down.

- **Life Expectancy vs. HDI ($r = 0.939$):** A tight, strong positive relationship. Countries with longer life expectancy (Sri Lanka, Bangladesh near the top) cluster closely around the regression line, while lower-life-expectancy countries sit consistently lower on HDI.
Interpretation: For South Asia, life expectancy is the more dependable predictor of HDI — the points hug the line closely. The gender-metric relationship is a bit noisier (more scatter around the line) but still strongly tied to HDI, just in the inverse direction: lower inequality coincides with higher development.
Middle East: Gender Development vs. HDI vs. Life Expectancy vs. HDI
- **Gender Development vs. HDI ($r = -0.976$):** An even steeper and tighter negative correlation than South Asia's — almost every point sits right on the line. Again, this points to the x-axis being an inequality-type measure rather than a parity measure, since the relationship is so strongly inverse.
- **Life Expectancy vs. HDI ($r = 0.933$):** Strong positive correlation, comparable in strength to South Asia's, though with slightly more scatter (a few mid-range life-expectancy countries sit a bit off the line).
Interpretation: In the Middle East, the gender-related metric is the single strongest predictor of HDI in either region (r close to -0.98) — countries with low inequality scores cluster at the top of the HDI range, while a couple of outliers near the bottom (low life expectancy *and* high inequality score) pull both regressions down. Life expectancy remains a solid secondary predictor, but slightly less tight than the gender metric.
One flag worth checking: since both "Gender Development" panels show negative correlations with HDI, it's worth double-checking what that variable actually represents in your dataset (GDI vs. GII, or however it was labeled/scaled) — the sign of the relationship changes the story you'd tell about it, and right now the data is describing an inequality-style metric, not a parity-style one.

Outlier Detection: HDI vs GNI per Capita (South Asia & Middle East, 2020–2022)



Outlier Detection (HDI vs. GNI per Capita)

Insight:

- South Asia: No countries flagged — the region's HDI-to-income relationship is tight and linear; even Sri Lanka and Maldives, which sit higher on HDI relative to their income, don't deviate enough from the IQR bounds to register as statistical outliers.
- Middle East: Yemen is flagged in every year (2020–2022) — its HDI sits around 0.42–0.43 even though several lower-middle-income Middle Eastern states post HDI well above 0.6 at similar or lower GNI levels. Interpretation: South Asia's countries largely convert income into development at a similar rate, so no country breaks far from the pack. Yemen's gap is different in kind, not just degree — its HDI is roughly 0.2–0.3 points lower than what its income level would predict elsewhere in the region, consistent with the disruption a prolonged conflict causes to health and education systems independent of resource availability.

Variability Comparison (Range & Coefficient of Variation)

Insight:

- South Asia: Range = 0.321, CV $\approx$ 0.154
- Middle East: Range = 0.513, CV $\approx$ 0.187

Interpretation: The Middle East spans a much wider HDI band — from Yemen's low 0.42 to UAE's high 0.93 — and that spread holds up even after adjusting for its higher average HDI (the CV is still higher). South Asia's countries cluster more tightly together; even its lowest performer (Afghanistan, $\sim$ 0.46–0.49) isn't as far below its regional average as Yemen is below the Middle East's.

What's actually driving HDI in each region

Checking correlations directly against HDI:

- South Asia: life expectancy ($r \approx 0.94$) edges out gender development ($r \approx 0.87$) as the stronger HDI predictor.
- Middle East: life expectancy ($r \approx 0.91$) and gender development ($r \approx 0.91$) are nearly tied — and gender *inequality* specifically has the strongest single relationship of any variable I checked ($r \approx -0.93$), meaning reductions in gender inequality track HDI gains even more tightly than income or longevity do.

Wealth-vs-development: Qatar and UAE post some of the highest GNI per capita in the dataset ($>$ \$65,000) but don't lead the region on HDI by an equivalent margin — Israel posts a comparable or higher HDI at less than half that income, illustrating that income translates unevenly into development outcomes even among wealthy states.

Policy angle: South Asia's case for prioritizing health infrastructure holds up in the numbers; the Middle East's case rests more evenly between health *and* gender-equity investment, with the inequality metric arguably the single highest-leverage lever based on correlation strength alone.

One thing worth flagging in your underlying chart: the "Gender Development" axis in your very first scatter plot showed a *negative* correlation with HDI — but in this dataset, gender_development actually correlates *positively* with HDI (0.87–0.91), while it's gender_inequality that's negative (-0.88 to -0.93). That strongly suggests your original chart was plotting the inequality column under a "Gender Development" label. Worth correcting before this goes into a final report, since it changes the wording of every claim about that variable (e.g., "higher gender development $\rightarrow$ higher HDI," not "higher gender development $\rightarrow$ lower HDI").

Conclusion

Throughout all of the problems analyzed within this report, the similar conclusions regarding human development can be made. Firstly, the single year snapshot (Problem 1A) revealed substantial global inequalities, as very high-HDI countries were found to be located mostly in Europe and some parts of the Asia-Pacific, while countries with the lowest levels of HDI were located in South Asia and Sub-Saharan Africa. Secondly, the trend analysis (Problem 1B) revealed that the picture did not stay the same – due to the COVID-19 pandemic, there was the first time of decline in global HDI, caused mostly by decrease in life expectancy of people, while poorer countries were recovering slower compared to wealthy countries. Within South Asia (Problem 2), it turned out that life expectancy was the highest predictor of HDI, while income had the declining marginal effect from some point. For instance, Sri Lanka, Bhutan, and the Maldives received a HDI that was higher than their income predicted, while Afghanistan, Pakistan, and Nepal have the HDI that was lower compared to their income.

When comparing the South Asia region with the Middle East (Problem 3), it becomes even more evident that the former conclusion regarding the HDI in South Asia.

Recommendations

- South Asia should prioritise public investment in healthcare, maternal and child health, and nutrition, since life expectancy gains showed the strongest statistical association with HDI improvement in the region.
- Targeted development support should be directed toward Afghanistan, Pakistan, and Nepal, where HDI consistently falls short of what income level alone would predict, focusing on the health and education gaps that separate them from higher-performing neighbours.
- Policymakers should study and, where possible, replicate the development strategies of Sri Lanka, Bhutan, and the Maldives, which convert relatively limited income into comparatively high HDI outcomes.
- The Middle East should prioritise closing gender gaps in education, workforce participation, and political representation, as reductions in gender inequality showed the strongest correlation with HDI of any variable examined in the region.
- High-GNI states such as Qatar and the UAE should be encouraged to diversify economically away from a narrow reliance on resource wealth, so that income growth translates more directly into broad-based improvements in health and education.
- Yemen, and similarly conflict-affected states, should be treated as a distinct policy category requiring humanitarian and institutional rebuilding support rather than conventional income-based

development assistance, since its HDI shortfall reflects structural disruption rather than resource scarcity alone.

- Future analysis should clearly verify and label gender-related indicators (Gender Development Index versus Gender Inequality Index) before drawing conclusions, since the direction of the relationship with HDI reverses depending on which is used, and this distinction materially changes the policy implications drawn from the data.

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